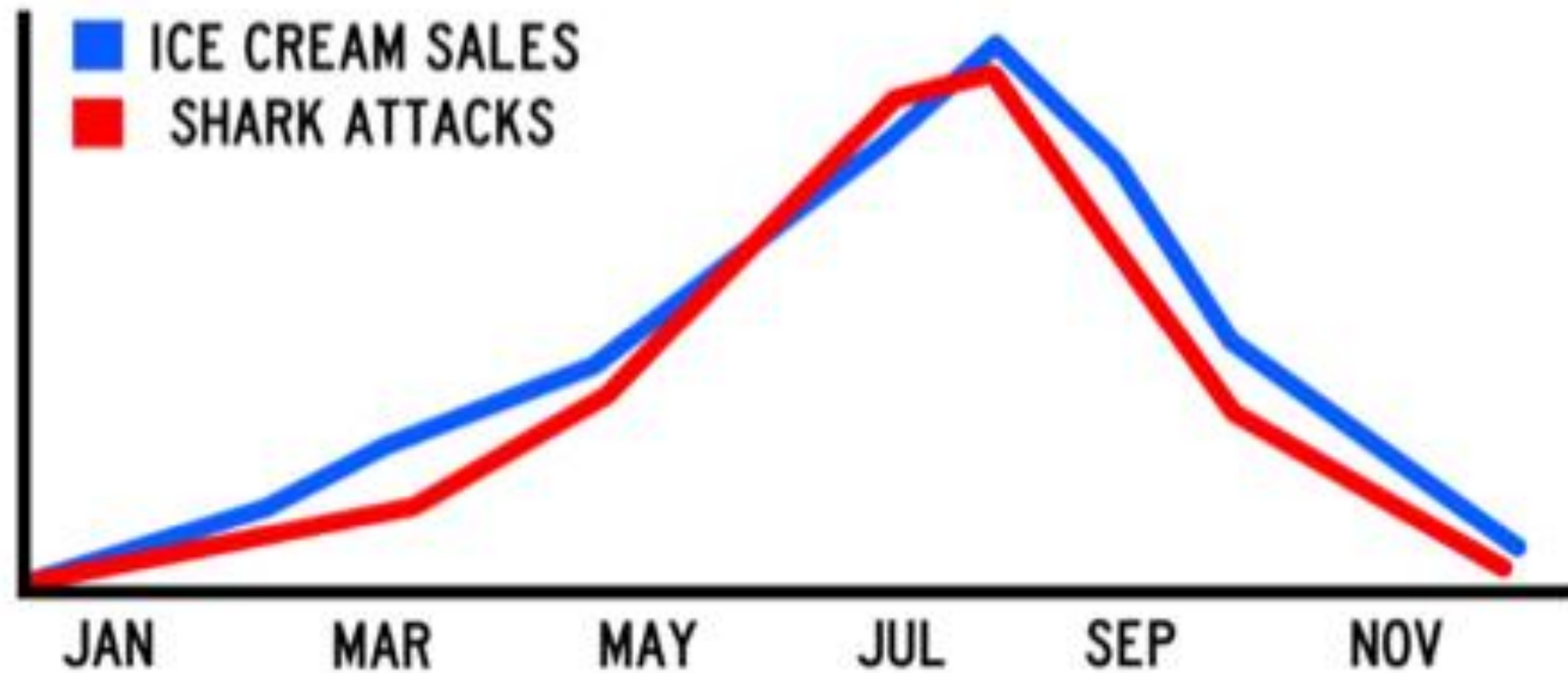


CORRELATION IS NOT CAUSATION!

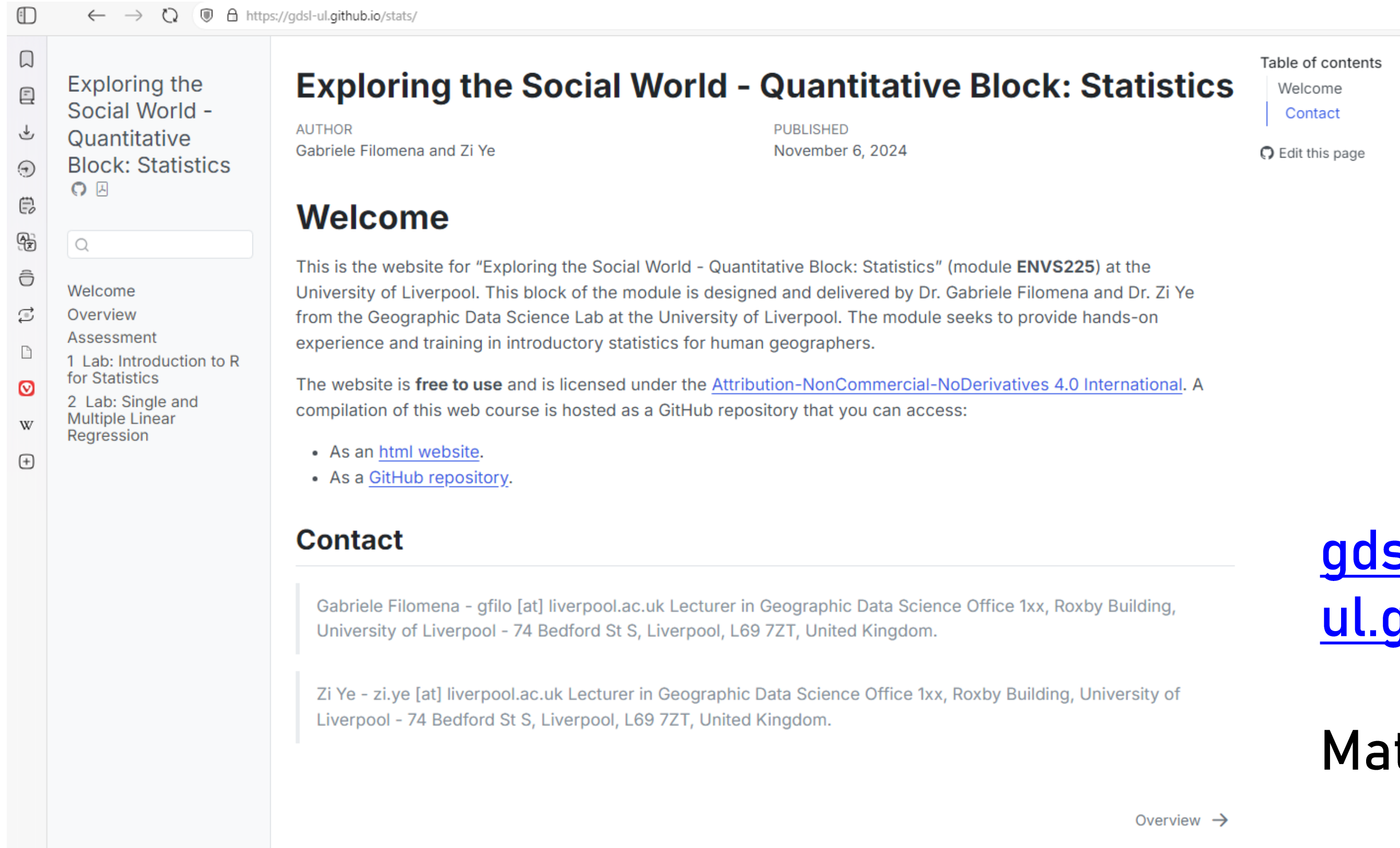


Intro & Review

ENVS225 Exploring the Social World

Gabriele Filomena
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Quantitative Data Block: **Webpage**



[gdsl-](https://github.com/gdsl)
[ul.github.io/stats/](https://github.com/gdsl)

Material and Labs

Quantitative Data Block: **Assignment**

Writing a research report using one of the regression techniques learned during the module. The report should have the following sections:

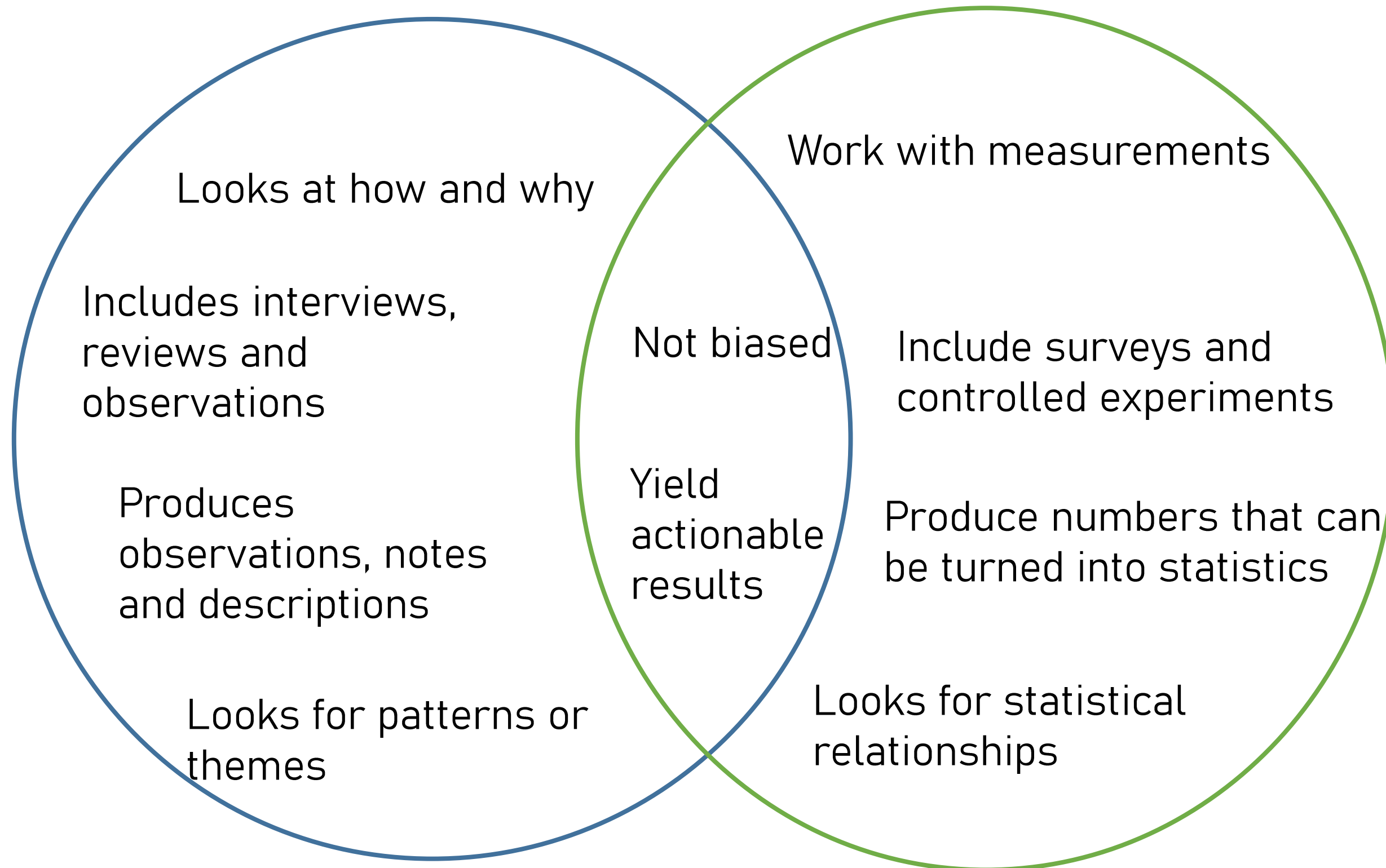
- An introduction.
- A literature review.
- Methods and data.
- Results and Discussion.
- Conclusion.

Deadline: Thursday 12 November 2026, 2pm - 2000 words - one .html file

See webpage. There is a Template

Qualitative

Quantitative



ENVS225 Stats – Ask for help?

- Ask questions on [Canvas](#)
- Come to class & labs!

Average Grade students who came to class: 67.5

Average Grade students who missed classes: 46.5

- Keep up with practical sessions. Do not underestimate this module!
- Stick to the instructions, do not make things harder for you.
- Office Hours 1-2pm on Friday 2nd, 9th, 30th October (Roxby Building, 1st floor Room 105).

Learning goals

- Understand how to explore a dataset, containing a number of observations described by a set of variables.
- Demonstrate an understanding in the application and interpretation of commonly used quantitative research methods.
- Demonstrate an understanding of how to work with quantitative data to address real-world research questions.

Structure

1	Introduction & Review	GF
2	Simple & Multiple Linear Regression	GF
3	Multiple Linear Regression with Categorical Variables	ZY
4	Logistic Regression	ZY
5	Data Visualisation	GF
6	Summary and Assessment Support	ZY

} + Support
Drop-in
sessions
(2h a week)

What are we going to work with?

nrKilled	nrWound	year	country	group	attack	target	weapon
0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades	Bombing/Explosion	Transportation	Explosives/Bombs/Dynamite
0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades	Bombing/Explosion	Transportation	Explosives/Bombs/Dynamite
0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades	Bombing/Explosion	Transportation	Explosives/Bombs/Dynamite
0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades	Bombing/Explosion	Transportation	Explosives/Bombs/Dynamite
0	1	1982	United Kingdom	Abu Nidal Organization (ANO)	Assassination	Government (Diplomatic)	Firearms
0	0	2014	United Kingdom	Anarchists	Facility/Infrastructure Attack	Business	Incendiary
0	0	2014	United Kingdom	Anarchists	Facility/Infrastructure Attack	Business	Incendiary
0	0	2014	United Kingdom	Anarchists	Facility/Infrastructure Attack	Business	Incendiary
0	0	2014	United Kingdom	Anarchists	Facility/Infrastructure Attack	Private Citizens & Property	Incendiary
0	0	2014	United Kingdom	Anarchists	Facility/Infrastructure Attack	Police	Incendiary
0	0	1984	United Kingdom	Animal Liberation Front (ALF)	Unarmed Assault	Private Citizens & Property	Chemical
0	0	1988	United Kingdom	Animal Liberation Front (ALF)	Facility/Infrastructure Attack	Business	Unknown
0	0	1988	United Kingdom	Animal Liberation Front (ALF)	Facility/Infrastructure Attack	Business	Unknown
0	0	1988	United Kingdom	Animal Liberation Front (ALF)	Facility/Infrastructure Attack	Business	Incendiary
0	0	1988	United Kingdom	Animal Liberation Front (ALF)	Facility/Infrastructure Attack	Business	Incendiary
0	0	1988	United Kingdom	Animal Liberation Front (ALF)	Facility/Infrastructure Attack	Business	Incendiary
0	0	1988	United Kingdom	Animal Liberation Front (ALF)	Facility/Infrastructure Attack	Business	Incendiary

Descriptive Statistics

Descriptive statistics

- Help us summarise the characteristics of our data/observations
- They form the initial step in a statistical analysis
- We get information about the centrality (e.g., mean, median)
- We get information about the variability (e.g., variance, standard deviation)

Mean

- *Mean* is another word for average.
- Most commonly used statistic that tells us about the centre of a data set.
- It is the average of the numbers in your dataset.

The sum of all values divided by the number of observations.

Median

- It is the middle point of your ordered data.
- If one orders the observations:

1,3,5,8,10

The median is the number in the middle

- When the number of observations is even:

1,3,5,6,8,10

Average the mid values -> 5.5

Difference between Mean and Median

- Both are single values that represent the centre of a data set.
- But median is more resilient to **outliers**.

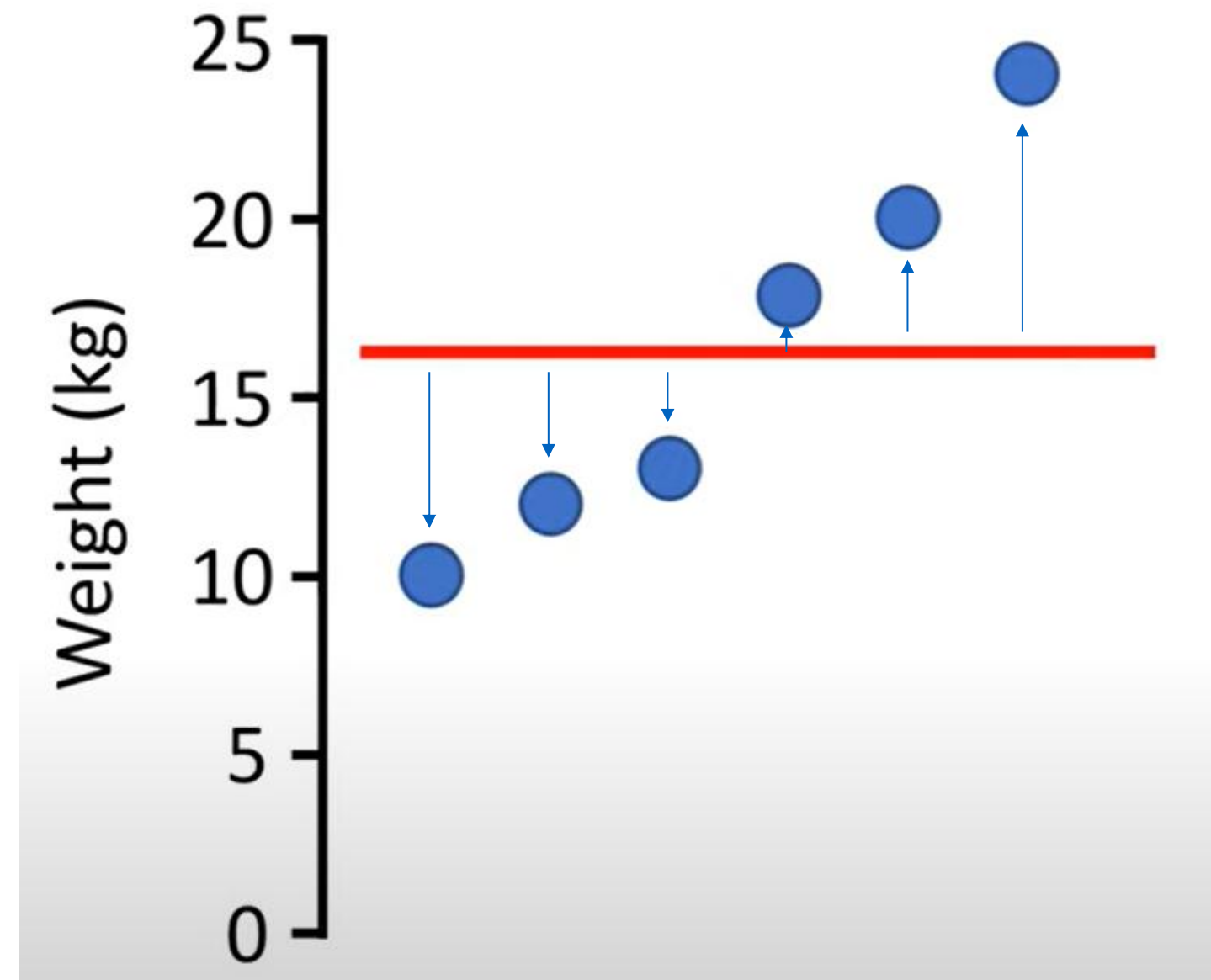
E.g.

3, 1, 4, 2.5, 50 (n=5)

- *Mean:* $\frac{3+1+4+2.5+50}{5}=12.1$
- *Median:* 1, 2.5, 3, 4, 50 =3

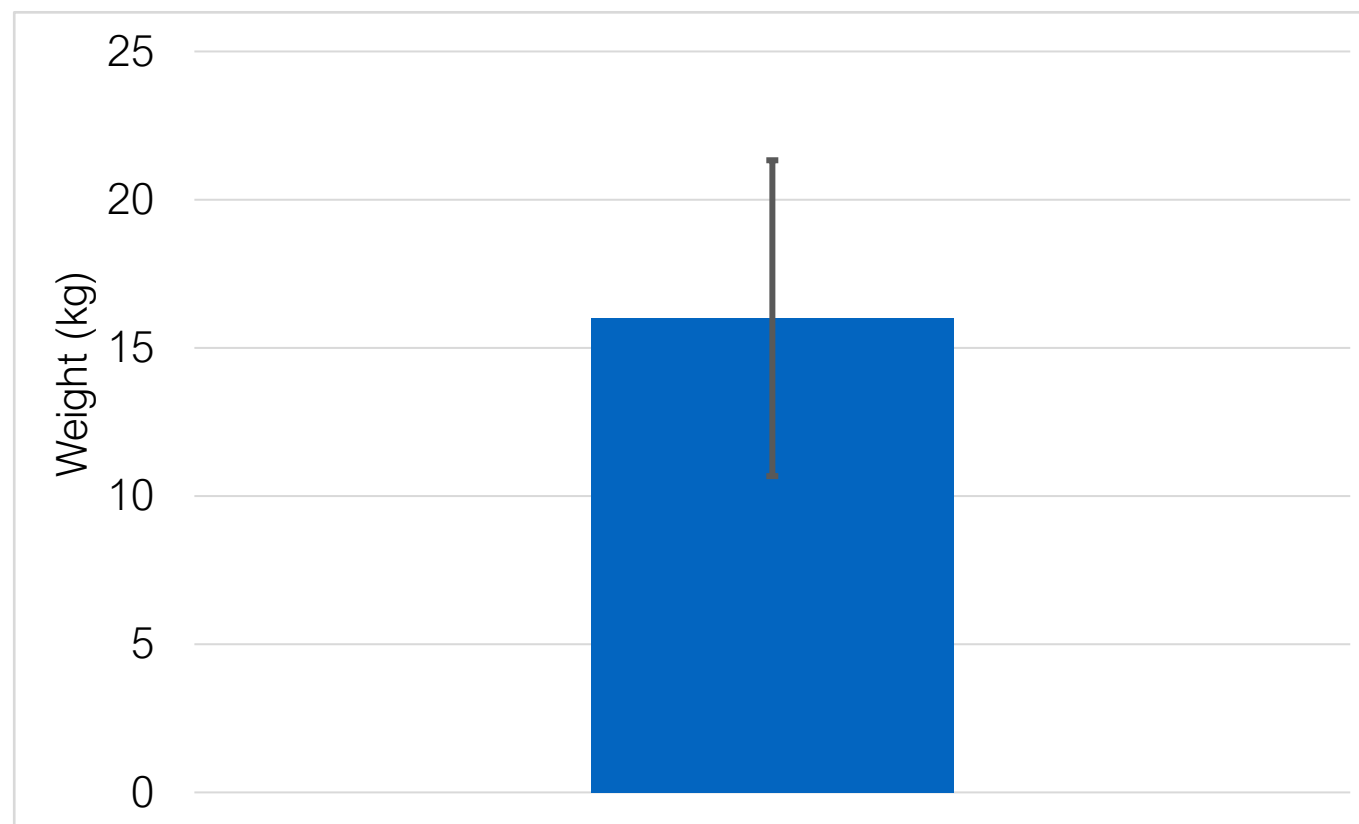
Variance (s^2)

- A measure to show how spread out our observations are
- Based on the distance of each observation from the mean
- The average of the squared differences from the mean.



Standard deviation (s)

- The square root of *variance*
- The result is in the same measurement units as our data (e.g., kg).
- This makes it easier to interpret and understand the spread of the data than *variance* alone.



- *Small value indicates data are gathered close to the mean*
- *Large value indicates data are spread far from the mean*

Activity:

Form groups of 5/6.

Find a dataset in a spreadsheet format (.csv, .xls)
Download, Open and Explore the dataset in MS Excel.

Be prepared to:

- describe the dataset, two variables. One qual, one quant.
- Find the mean and the standard deviation of the quant variable.

Associations between Variables

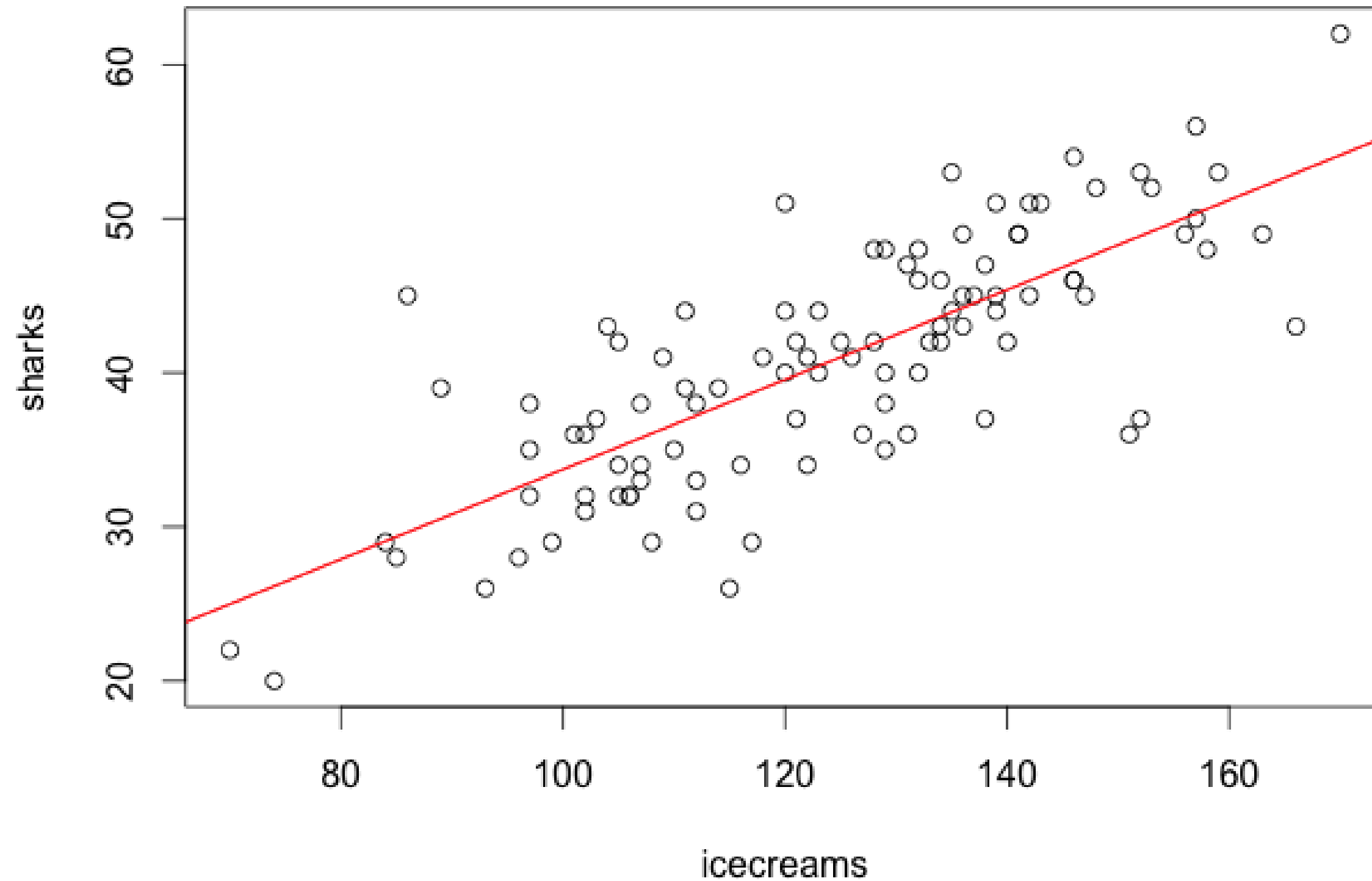
Correlation

- Describes an association between variables.
- When one variable changes, so does the other.
- **Covariation**, rather than a direct causal link.

Examples:

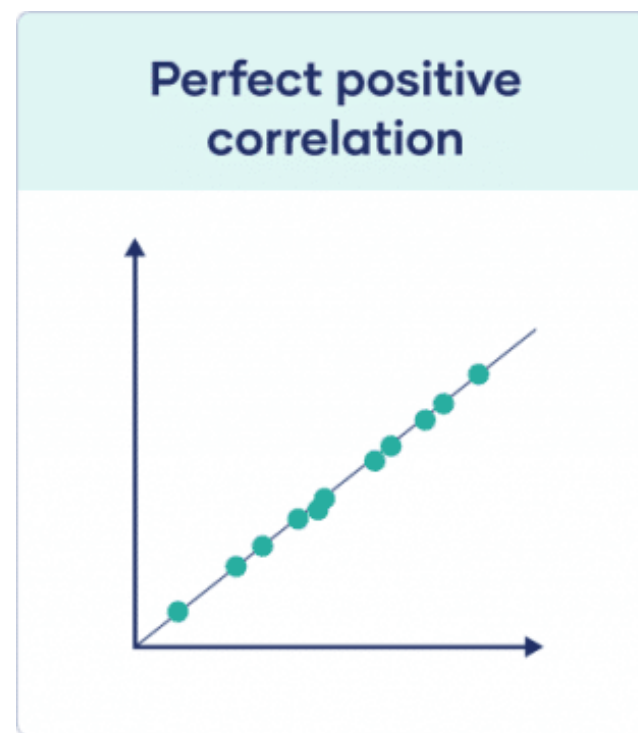
- Ice cream sales and shark attacks.
- Time spent in shops and temperature.

What is Correlation?



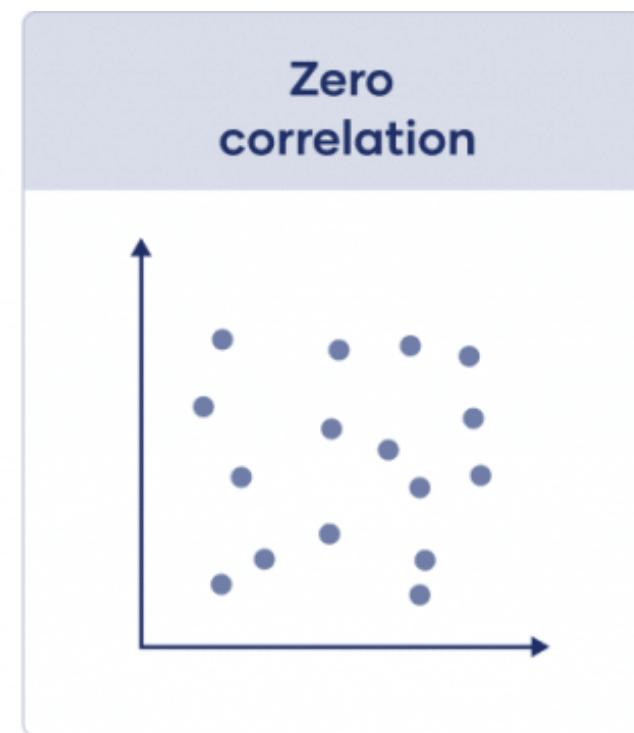
Pearson Correlation

The Pearson correlation coefficient (r) measures the strength and direction of the linear relationship between **continuous** variables.

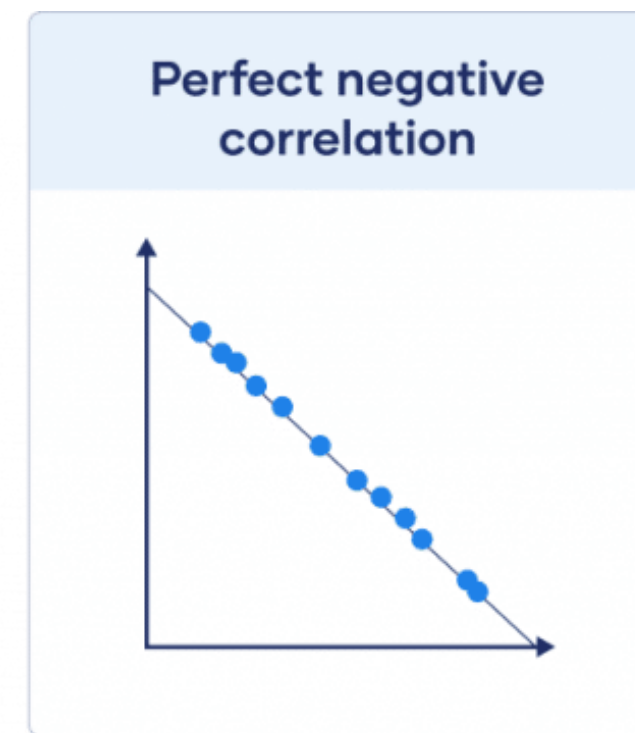


$$r = 1$$

Same direction



$$r = 0$$



$$r = -1$$

*Opposite
direction*

Simple Linear Regression

- Regression is used to estimate the relationship between two variables.
- How a change in one variable (e.g., someone's exercise habits) can predict the outcome of another variable (e.g., general health).

DEPENDENT ◀ INDEPENDENT variables

=

RESPONSE ◀ EXPLANATORY

Examples

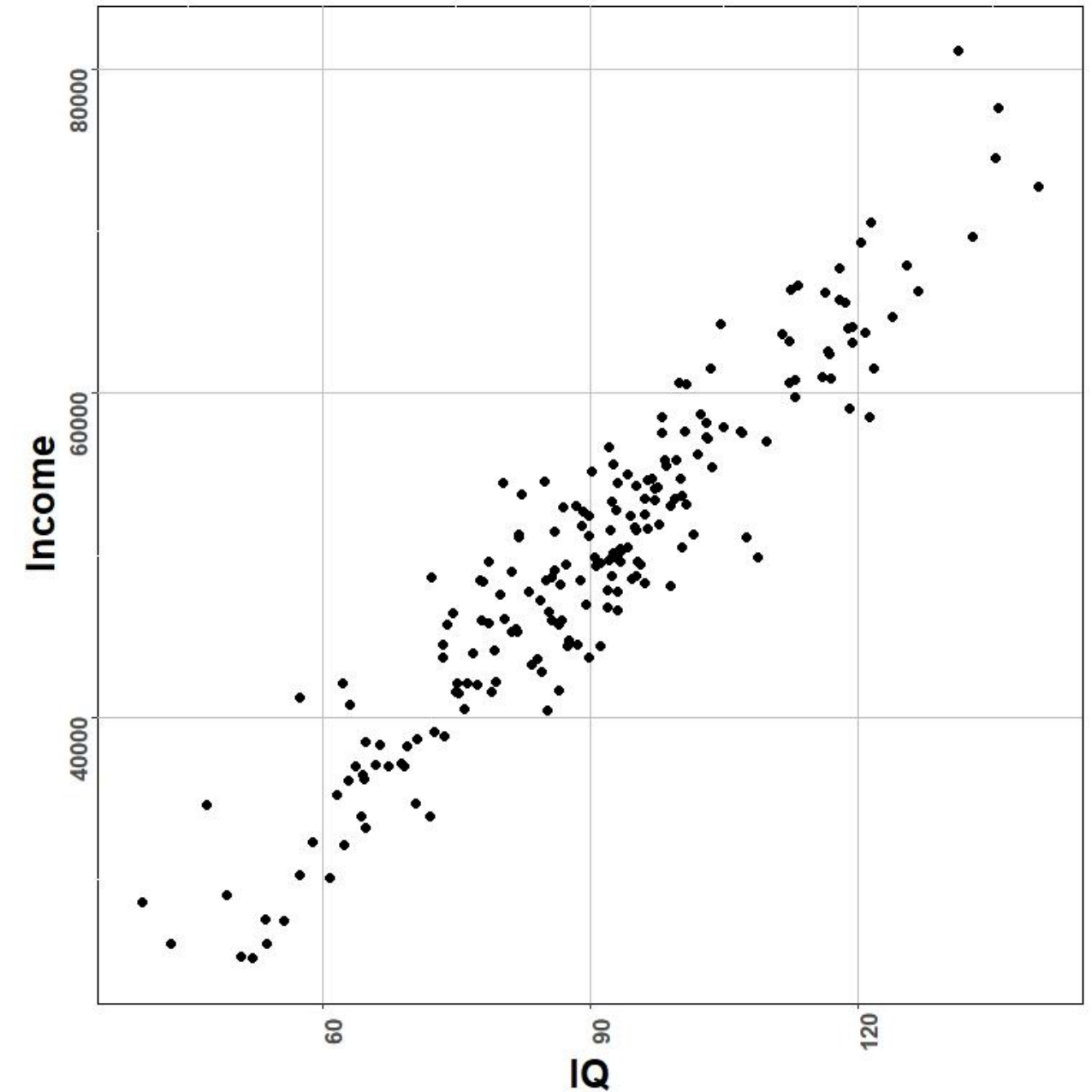
- IQ and income (IQ is the [REDACTED] and income is the [REDACTED])?
- Study time and grades (hours of study time is the [REDACTED] and average grade is the [REDACTED])?
- Exercise and blood pressure (hours of exercise is the [REDACTED] and blood pressure is the [REDACTED])?

Examples

- IQ and income (IQ is the **INDEPENDENT** and income is the **DEPENDENT**)?
- Study time and grades (hours of study time is the **INDEPENDENT** and average grade is the **DEPENDENT**)?
- Exercise and blood pressure (hours of exercise is the **INDEPENDENT** and blood pressure is the **DEPENDENT**)?

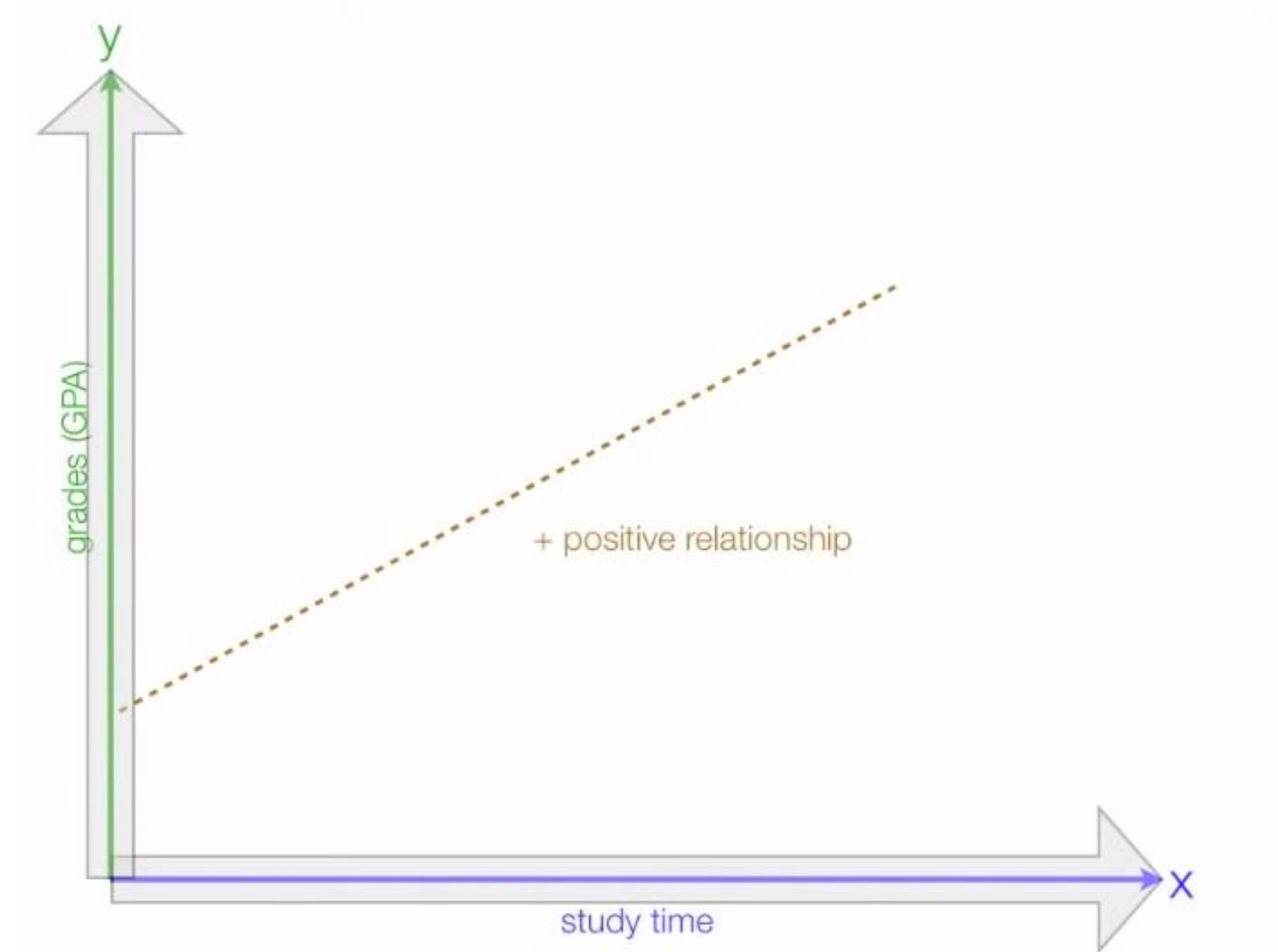
Displaying the data

When both the **DEPENDENT** and **INDEPENDENT** are numerical, we can represent data in the form of a scatterplot.



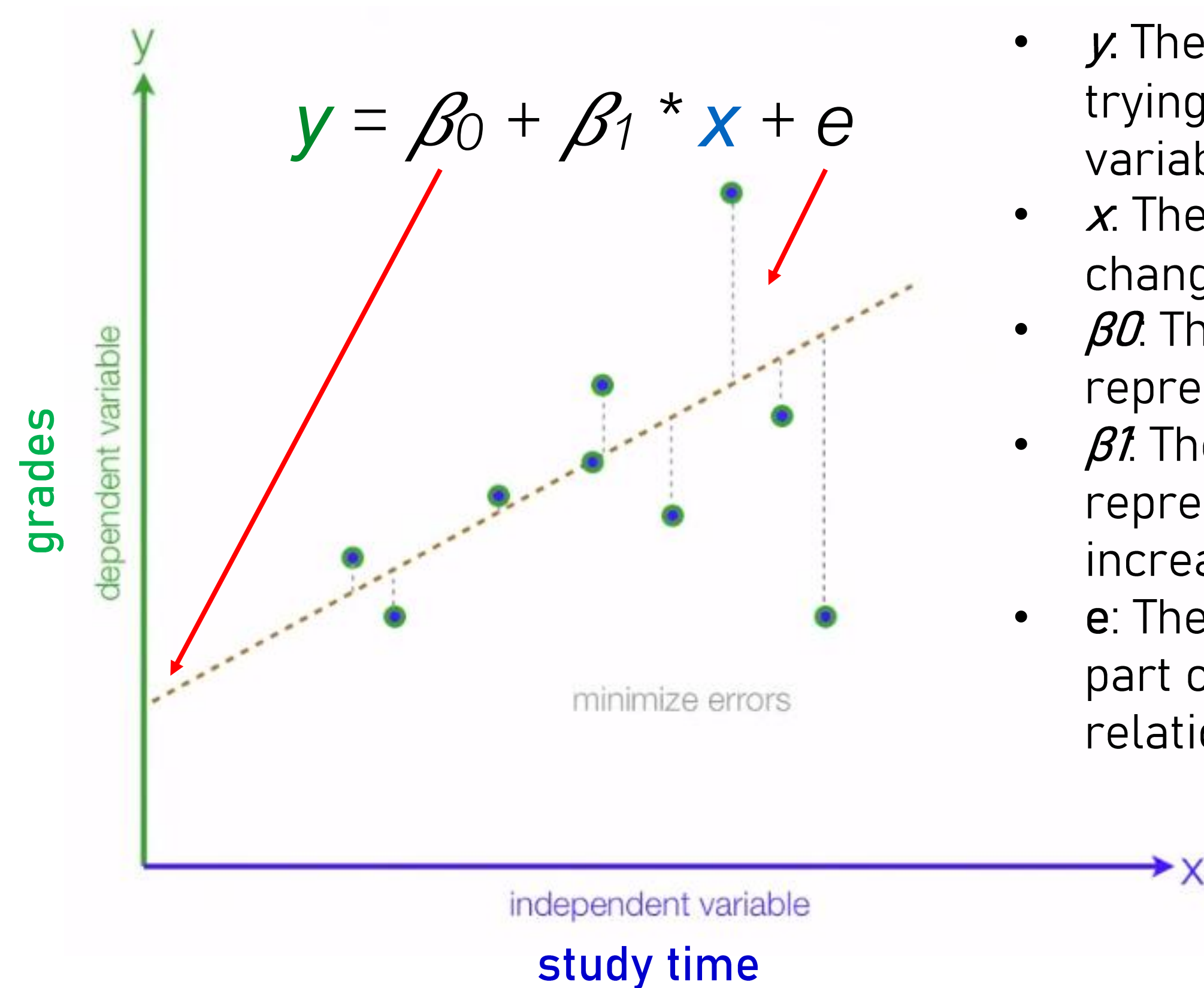
Simple Linear Regression

- Straight line prediction model.
- As an **INDEPENDENT** variable changes, what happens to the **DEPENDENT** variable?



Correlation vs Simple Linear Regression

- **Correlation** quantifies the strength and direction of the linear relationship between two variables but doesn't give information about the equation of the line.
- **Simple Linear Regression** goes further by providing an equation that describes the linear relationship.
- **Simple Linear Regression** models the relationship between two variables.



- y : The dependent variable. The outcome we are trying to explain based on the independent variable.
- x : The independent variable used to explain changes in the dependent variable.
- β_0 : The intercept of the regression line. It represents the expected value of y when x is 0.
- β_1 : The slope of the regression line. It represents the change in y for a one-unit increase in x .
- e : The error term (or residual). It represents the part of y that cannot be explained by the linear relationship with x .

